Empathy in Real vs. Synthetic Clinician Comments

0.1 Summary

Across 3 synthetic models (Gemini, Grok, Openai) and 3 prompting methods (DG, SAG, SAGII), real comments generally show significantly higher empathy ratings across all 3 mechanisms (Emotional Reactions, Interpretations, Explorations), with one exception: for Interpretations, Grok scores significantly *higher* than real across all 3 prompting strategies (DG, SAG, SAGII).

0.2 Data

6600 Reddit posts were used, each paired with its real clinician comment(s) and synthetic comments from 3 models x 3 prompting methods. 137278 comments were classified in total: 8625 real and 128653 synthetic (14295 per model/method group on average). Predicted empathy comes from the same three fine-tuned RoBERTa bi-encoder classifiers used in pilot 1 (Emotional Reactions, Interpretations, Explorations; see `../2_training`), each producing an ordinal label: 0 = none, 1 = weak, 2 = strong.

0.3 Mean predicted empathy by group

Empathy labels are ordinal (0 = none, 1 = weak, 2 = strong). The means shown below implicitly treat the labels as equally spaced, and are shown only as a quick, monotonic summary (Fig. 1). The actual label distribution (the ordinal-honest view) is plotted underneath it as proportions per group (Fig. 2).

0.4 Real vs. synthetic (Mann-Whitney U, Benjamini-Hochberg adjusted)

Table 2 shows that all real-vs-synthetic contrasts are significant, such that real ones score significantly higher. Notable exceptions, where the synthetic ones score higher are marked in red.

0.5 Mixed-effects models

The Mann-Whitney tests above treat every comment as independent, but they aren't: several synthetic comments (and sometimes several real ones) come from the *same* Reddit thread, so they share whatever makes that particular topic more or less emotionally charged or explorable. To account for that, we run two mixed-effects models per mechanism:

- **A - omnibus** (`label ~ group + log(length) + (1|post_id)`, all 137278 comments): tests whether the 10-level group factor (real + 9 model/method combinations) matters at all.
- **B - model × method** (`label ~ model * method + log(length) + (1|post_id)`, synthetic comments only - "real" has no model/method to cross): tests the model main effect, the method main effect, and whether they interact.

Emotional Reactions uses all 3 empathy levels, so it's fit as a cumulative link mixed model (`ordinal::clmm`). **Interpretations** and **Explorations** never predict label 1 anywhere in this data (0 occurrences out of 137278 comments, checked directly), so a 3-level ordinal model would be degenerate for them—they're fit instead as binomial mixed models (`lme4::glmer`) on P(label = 2).

Emotional Reactions's Model A (omnibus test, vs.-real contrasts) had a small residual gradient (max|grad| = 0.0147 (didn't fully settle)) but its estimates are usable and reported below as directionally informative. **Model B (the model x method interaction) failed to converge outright** (max|grad| = 280, Hessian is not usable - fit did not converge, results not reliable) - its Hessian isn't usable, so no valid significance test or simple-effects contrasts exist for it, and none are reported below; this is plausibly related

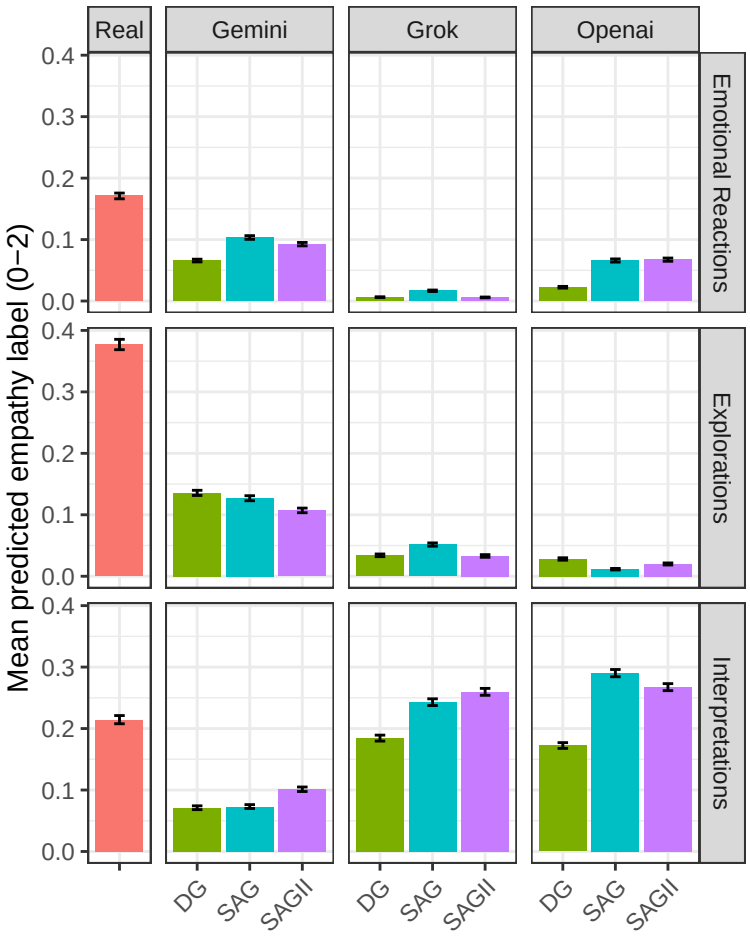


Figure 1: Average predicted empathy per population.

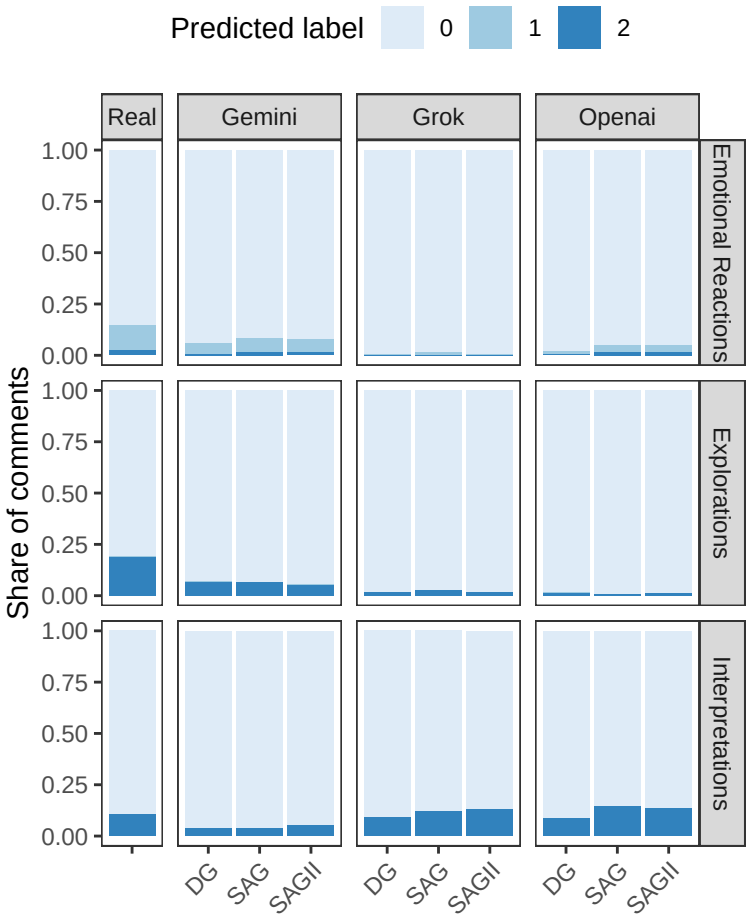


Figure 2: Stacked predicted label proportions per population.

Table 1: Predicted label proportions by group (same data as Fig. 2, as a table)—useful for spotting cells too sparse to trust, e.g. where P(label=1) is 0.

Mechanism	Model	Method	n	P(label=0)	P(label=1)	P(label=2)
Emotional Reactions	Real		8625	0.853	0.123	0.024
	Gemini	DG	14301	0.939	0.056	0.005
		SAG	14288	0.914	0.068	0.018
		SAGII	14300	0.921	0.065	0.014
	Grok	DG	14258	0.994	0.006	0.000
		SAG	14300	0.985	0.013	0.002
		SAGII	14300	0.994	0.005	0.000
	Openai	DG	14302	0.981	0.016	0.003
		SAG	14302	0.951	0.032	0.017
		SAGII	14302	0.950	0.033	0.017
Explorations	Real		8625	0.811	0.000	0.189
	Gemini	DG	14301	0.932	0.000	0.068
		SAG	14288	0.937	0.000	0.063
		SAGII	14300	0.946	0.000	0.054
	Grok	DG	14258	0.983	0.000	0.017
		SAG	14300	0.974	0.000	0.026
		SAGII	14300	0.983	0.000	0.017
	Openai	DG	14302	0.986	0.000	0.014
		SAG	14302	0.994	0.000	0.006
		SAGII	14302	0.990	0.000	0.010
Interpretations	Real		8625	0.893	0.000	0.107
	Gemini	DG	14301	0.964	0.000	0.036
		SAG	14288	0.963	0.000	0.037
		SAGII	14300	0.949	0.000	0.051
	Grok	DG	14258	0.908	0.000	0.092
		SAG	14300	0.879	0.000	0.121
		SAGII	14300	0.870	0.000	0.130
	Openai	DG	14302	0.914	0.000	0.086
		SAG	14302	0.855	0.000	0.145
		SAGII	14302	0.866	0.000	0.134

Table 2: Mann-Whitney U test.

Mechanism	Model	Method	Mean (real)	Mean (synthetic)	p	p (BH-adj.)	Rank-biserial r
Emotional Reactions	Gemini	DG	0.171	0.066	0.0000	0.0000	0.087
		SAG	0.171	0.103	0.0000	0.0000	0.060
		SAGII	0.171	0.093	0.0000	0.0000	0.068
	Grok	DG	0.171	0.006	0.0000	0.0000	0.141
		SAG	0.171	0.017	0.0000	0.0000	0.132
		SAGII	0.171	0.006	0.0000	0.0000	0.141
	Openai	DG	0.171	0.022	0.0000	0.0000	0.127
		SAG	0.171	0.066	0.0000	0.0000	0.097
		SAGII	0.171	0.067	0.0000	0.0000	0.096
Explorations	Gemini	DG	0.377	0.136	0.0000	0.0000	0.121
		SAG	0.377	0.127	0.0000	0.0000	0.125
		SAGII	0.377	0.107	0.0000	0.0000	0.135
	Grok	DG	0.377	0.034	0.0000	0.0000	0.171
		SAG	0.377	0.052	0.0000	0.0000	0.163
		SAGII	0.377	0.033	0.0000	0.0000	0.172
	Openai	DG	0.377	0.028	0.0000	0.0000	0.174
		SAG	0.377	0.011	0.0000	0.0000	0.183
		SAGII	0.377	0.020	0.0000	0.0000	0.179
Interpretations	Gemini	DG	0.214	0.071	0.0000	0.0000	0.072
		SAG	0.214	0.073	0.0000	0.0000	0.071
		SAGII	0.214	0.101	0.0000	0.0000	0.057
	Grok	DG	0.214	0.184	0.0002	0.0002	0.015
		SAG	0.214	0.243	0.0011	0.0011	-0.014
		SAGII	0.214	0.260	0.0000	0.0000	-0.023
	Openai	DG	0.214	0.172	0.0000	0.0000	0.021
		SAG	0.214	0.290	0.0000	0.0000	-0.038
		SAGII	0.214	0.267	0.0000	0.0000	-0.026

Table 3: Convergence status for each fitted model.

Mechanism	Model	Type	Converged	Note
Emotional Reactions	A_group_omnibus	clmm	FALSE	max|grad| = 0.0147 (didn't fully settle)
Emotional Reactions	B_model_x_method	clmm	FALSE	max|grad| = 280, Hessian is not usable - fit did not c
Interpretations	A_group_omnibus	glmer	TRUE	NA
Interpretations	B_model_x_method	glmer	TRUE	NA
Explorations	A_group_omnibus	glmer	TRUE	NA
Explorations	B_model_x_method	glmer	TRUE	NA

Table 4: Model A: omnibus test for group (real + 9 model/method combinations), replacing Kruskal-Wallis.

Mechanism	Chi-sq	df	p
Emotional Reactions	3996.9	9	0
Interpretations	2603.3	9	0
Explorations	4464.7	9	0

to how sparse label 2 is for the thinnest group (grok_dg: 2/14258 at label 2). IP and EX converged cleanly throughout.

The group effect is significant for all 3 mechanisms (Table 4). Comment length (log-transformed) is included as a covariate in both models and is a significant predictor everywhere (Model A: all 3 $p < 0.001$), but not in the same direction: Emotional Reactions, Interpretations score higher with more words, while Explorations scores lower with more words—plausibly because exploratory, question-asking replies tend to be short follow-ups rather than long ones.

Table 5: real scores significantly higher than every synthetic group for Emotional Reactions, Explorations; Interpretations is the exception—grok_dg, grok_sag, grok_sagii score significantly *higher* than real.

This isn't quite what the raw Mann-Whitney table shows: refitting without `log(length)` flips the direction for grok_dg, openai_sag, openai_sagii (3 groups), reproducing the raw pattern for those groups almost exactly—so this looks like a length confound: comment length differs systematically across models and strongly predicts this label, so part of the length-adjusted picture for these groups is attributable to comment length rather than the model/method itself.

The model x method interaction is significant for all 2 mechanisms it could be tested for—not estimable for Emotional Reactions, whose Model B fit failed to converge, i.e. for those mechanisms the effect of a prompting method genuinely depends on which model generated it. Both main effects (model and method) are significant for every mechanism they could be tested for, on top of the interaction. Emotional Reactions's model and method main effects aren't estimable here—its Model B fit failed to converge, so there's no valid test for any term in it beyond what's noted above.

Since the interaction is significant for at least one mechanism, pooled main-effect means (averaged over the other factor) would be misleading there - e.g. a model's "average" effect can look muted if it does well with one method and poorly with another. So instead, two sets of simple-effects contrasts throughout: models compared within each method, and methods compared within each model. Emotional Reactions is omitted from both tables below—its Model B fit failed to converge, so there are no valid contrasts to report.

Table 5: Model A: each (model, method) group vs. real (OR below 1 = lower odds of higher empathy than real; rows with OR above 1 shaded).

Mechanism	Group	Odds ratio	OR 95% LCL	OR 95% UCL	p (BH-adj.)
Emotional Reactions	gemini_dg	0.279	0.251	0.311	0
Emotional Reactions	openai_dg	0.052	0.045	0.060	0
Emotional Reactions	grok_dg	0.021	0.017	0.027	0
Emotional Reactions	gemini_sag	0.431	0.391	0.476	0
Emotional Reactions	openai_sag	0.128	0.114	0.144	0
Emotional Reactions	grok_sag	0.048	0.041	0.056	0
Emotional Reactions	gemini_sagii	0.359	0.325	0.396	0
Emotional Reactions	openai_sagii	0.133	0.118	0.149	0
Emotional Reactions	grok_sagii	0.016	0.013	0.021	0
Interpretations	gemini_dg	0.531	0.464	0.606	0
Interpretations	openai_dg	0.604	0.544	0.671	0
Interpretations	grok_dg	2.062	1.832	2.320	0
Interpretations	gemini_sag	0.475	0.416	0.541	0
Interpretations	openai_sag	0.648	0.586	0.716	0
Interpretations	grok_sag	2.041	1.830	2.277	0
Interpretations	gemini_sagii	0.594	0.526	0.670	0
Interpretations	openai_sagii	0.578	0.523	0.639	0
Interpretations	grok_sagii	2.453	2.197	2.739	0
Explorations	gemini_dg	0.195	0.177	0.216	0
Explorations	openai_dg	0.036	0.031	0.043	0
Explorations	grok_dg	0.036	0.031	0.042	0
Explorations	gemini_sag	0.185	0.167	0.205	0
Explorations	openai_sag	0.017	0.013	0.022	0
Explorations	grok_sag	0.063	0.055	0.071	0
Explorations	gemini_sagii	0.155	0.139	0.173	0
Explorations	openai_sagii	0.030	0.025	0.037	0
Explorations	grok_sagii	0.037	0.032	0.043	0

Table 6: Model B: model, method, and model x method tests (synthetic comments only), replacing the pooled by-method/by-model comparisons.

Mechanism	Term	Chi-sq	df	p
Emotional Reactions	model	NA	NA	NA
Emotional Reactions	method	NA	NA	NA
Emotional Reactions	model:method	NA	NA	NA
Interpretations	model	2742.7	2	0.0000
Interpretations	method	11.3	2	0.0035
Interpretations	model:method	47.3	4	0.0000
Explorations	model	1378.5	2	0.0000
Explorations	method	40.9	2	0.0000
Explorations	model:method	101.3	4	0.0000

Table 7: Model B: pairwise model contrasts within each method.

Mechanism	Method	Contrast	Odds ratio	OR 95% LCL	OR 95% UCL	p (BH-adj.)
Interpretations	dg	gemini / openai	1.202	1.059	1.364	0.0051
Interpretations	dg	gemini / grok	0.237	0.212	0.265	0.0000
Interpretations	dg	openai / grok	0.197	0.176	0.221	0.0000
Interpretations	sag	gemini / openai	1.116	0.980	1.271	0.1026
Interpretations	sag	gemini / grok	0.228	0.204	0.254	0.0000
Interpretations	sag	openai / grok	0.204	0.183	0.227	0.0000
Interpretations	sagii	gemini / openai	1.498	1.333	1.684	0.0000
Interpretations	sagii	gemini / grok	0.218	0.198	0.241	0.0000
Interpretations	sagii	openai / grok	0.146	0.130	0.163	0.0000
Explorations	dg	gemini / openai	3.478	2.916	4.147	0.0000
Explorations	dg	gemini / grok	7.986	6.801	9.377	0.0000
Explorations	dg	openai / grok	2.296	1.846	2.857	0.0000
Explorations	sag	gemini / openai	3.827	2.901	5.047	0.0000
Explorations	sag	gemini / grok	3.327	2.892	3.827	0.0000
Explorations	sag	openai / grok	0.869	0.650	1.163	0.3455
Explorations	sagii	gemini / openai	1.933	1.537	2.430	0.0000
Explorations	sagii	gemini / grok	5.782	4.895	6.829	0.0000
Explorations	sagii	openai / grok	2.992	2.290	3.910	0.0000

Table 8: Model B: pairwise method contrasts within each model.

Mechanism	Model	Contrast	Odds ratio	OR 95% LCL	OR 95% UCL	p (BH-adj.)
Interpretations	gemini	dg / sag	1.161	1.018	1.323	0.0390
Interpretations	gemini	dg / sagii	0.971	0.858	1.099	0.7001
Interpretations	gemini	sag / sagii	0.837	0.741	0.945	0.0084
Interpretations	openai	dg / sag	1.078	0.988	1.176	0.1174
Interpretations	openai	dg / sagii	1.211	1.109	1.322	0.0001
Interpretations	openai	sag / sagii	1.123	1.042	1.210	0.0051
Interpretations	grok	dg / sag	1.115	1.023	1.216	0.0223
Interpretations	grok	dg / sagii	0.894	0.821	0.973	0.0165
Interpretations	grok	sag / sagii	0.802	0.741	0.867	0.0000
Explorations	gemini	dg / sag	0.904	0.808	1.012	0.1106
Explorations	gemini	dg / sagii	0.974	0.864	1.097	0.7001
Explorations	gemini	sag / sagii	1.077	0.958	1.210	0.2590
Explorations	openai	dg / sag	0.995	0.745	1.329	0.9740
Explorations	openai	dg / sagii	0.541	0.422	0.694	0.0000
Explorations	openai	sag / sagii	0.544	0.408	0.724	0.0001
Explorations	grok	dg / sag	0.377	0.314	0.453	0.0000
Explorations	grok	dg / sagii	0.705	0.578	0.860	0.0014
Explorations	grok	sag / sagii	1.871	1.562	2.241	0.0000